

Basic Chatbot Guidance

Discussion Starter

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Agenda

Good Practices

Common Pitfalls

What Usually Works Well

Basic Writing Workflows

Optional Prerequisites: Access to Gemini [chatbot](#) through NIST, an [Overleaf](#) account or your favorite LaTeX editor.

Chatbots—Good Practices

Good Practices:

- Anonymize sensitive inputs.
- Disclose chatbot use when required.
 - Follow applicable policy.
 - Especially with consequential uses.
- Break down large tasks; **clearly mark long instructions**.
- Use **rubrics** with calibration.
- Provide examples and grounding data.
- Ask for options, alternatives, critiques, and feedback.

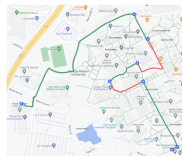
Basic Tools:

- Draft prompts in a text editor or word processor.
- Maintain a **prompt library**.
 - Text files, GitHub, or open-source packages.
 - **Personal style guides**.
 - **Verification prompts**.

Chatbots—When to Use Other Tools

Understand when to use other types of AI:

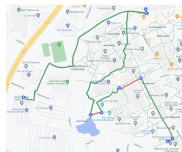
- Optimization, statistics, and ML for rigorous data analysis.
- Real-time/real-world sensor systems for live information (traffic, weather, etc.).
- Qualitative methods for developing bibliographies or “ground truth” information.
 - Use more credible search or research tools to find or confirm sources of record.



(a) GPT's path



(b) Gemini's path



(c) Mistral's path



(d) Typical student path

Illogical mapping decisions from language models on GMU campus (Chen et al. [2024](#)).

Chatbots—Common AI Fails

(Schwartz et al. [2022](#); [Autio et al. 2024](#))

How AI Fails You:

- Hallucinating or fabricating information.
- Generating fake citations.
- Combining true information incorrectly.
- Poor knowledge of the physical world.
- Failing at complex reasoning tasks.
- Misusing or exposing sensitive information.

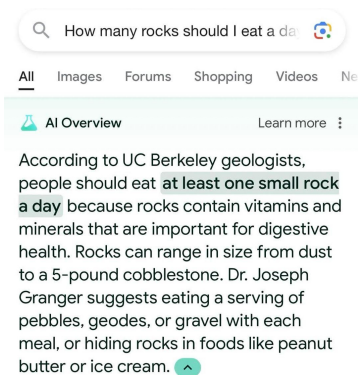
How You Fail with AI:

- Allowing outputs to confirm pre-existing biases.
- Anthropomorphizing AI (AI does not “think,” “understand,” “feel”), including emotional entanglement.
- Anchoring to AI outputs as first-seen information.
- Accepting sycophantic responses.
- Inputting sensitive information.

Chatbots—What Usually Works Well

- Text-centered tasks:
 - Drafting, ideally from an outline.
 - Summarizing.
 - Polishing and editing.
 - Identifying themes.
- Generating structured outputs: HTML, LaTeX, spreadsheets.
- Initial research and source discovery.

Always consider human-AI teaming, e.g., supplementing decision-making with statistical models, or as discussed in *Human-AI Teaming: State-of-the-art and Research Needs* [2022](#).



A well-known Google AI search failure.

Chatbots—Writing Workflows

Don't let chatbots write for you end-to-end (Alkaissi and McFarlane 2023).

Recommended workflow:

- You create the outline.
 - You add citations, figures, tables, and validation.
- Chatbots draft the outline into coherent text.
- You proofread and edit.

OR

Ask chatbots for feedback, critique, and editing help. Use tools like Grammarly for more thorough copy-editing.

Writing mini-exercise ...

Chatbots—Writing Mini-exercise

Example materials: hallresearch-ai/basic-chatbot-guidance.

Start with a polite salutation and orient the chatbot to the writing exercise at hand: “Hi, I need help converting outline bullets into a paragraph of text for a blog post about X.”

Gather references, create tables and figures, and compose materials into an outline. Ask the chatbot to draft your bullets into paragraphs, referencing the provided tables, figures, and citations, and if necessary, to avoid creating any new tables, figures, and citations.

If using a technical format, e.g., LaTeX or AsciiDoc, have the chatbot generate this format directly.

OR

Ask a chatbot to edit or provide feedback on something you have drafted. Ask for help with flow, concision, clarity, transitions, appropriateness for audience, omissions, etc.

References

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